

2022

Time : $1\frac{1}{2}$ hours

Full Marks : 40

*Candidates are required to give their answers
in their own words as far as practicable.*

The figures in the margin indicate full marks.

Answer from both the sections as directed.

Section–A

Answer any **four** questions of the following :

$$6 \times 4 = 24$$

1. Prove that the identity element in a group is unique.

2. Let G be a group and H a non-empty subset of G . Then H is a subgroup of G if and only if $a \in H, b \in H \Rightarrow a.b^{-1} \in H$.
3. To find the necessary and sufficient condition for a non-empty subset N of a vector space $V(F)$ to be a subspace of $V(F)$ is $a, b \in F$ and $\alpha, \beta \in W \Rightarrow \alpha a + b\beta \in W$.
4. The intersection of any two subspaces of a vector space is a subspace.
5. If A, B be two non singular matrices of the same order then $(AB)^{-1} = B^{-1}A^{-1}$.

6. Find the inverse of the matrix

$$A = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 3 & 1 & 1 \end{bmatrix}$$

Section-B

Answer any **four** questions of the following :

$$4 \times 4 = 16$$

7. Prove that $G = \{1, -1, i, -i\}$ is a group with respect to multiplication.
8. Prove that every field is an integral domain but converse is not necessarily true.
9. Let $V(F)$ be a vector space and O be the zero vector of V . Then
 - (a) $a \cdot 0 = 0 \forall a \in F$
 - (b) $a \cdot a = b \cdot \beta \Rightarrow a = b \forall a, b \in F$ and $a \in V$ and $a \neq 0$.
10. Define vector subspace Show that the W of ordered triads $(a_1, a_2, 0)$ where $a_1, a_2 \in F$ is a subspace of $V_3(F)$.

11. Solve the following system of equation by matrix methods.

$$x - 2y + 3z = 5$$

$$4x - 3y + 4z = 7$$

$$x + y + z = -4$$

12. Find the Eigen value of the matrix

$$A = \begin{bmatrix} 9 & -1 & 9 \\ 3 & -1 & 3 \\ -7 & 1 & -7 \end{bmatrix}$$

$$\frac{\quad}{\quad} \times \frac{\quad}{\quad}$$